

MATH0109 Theorem Proving in Lean

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0109
<i>Level:</i>	6 (UG), 7 (PG)
<i>Normal student group(s):</i>	UG: Y3 Mathematics programmes
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	2
<i>Assessment:</i>	25% coursework, 40% departmental test(s), 35% written report. The departmental tests consist of two 20% tests.
<i>Normal Pre-requisites:</i>	Familiarity with the notion of proving theorems (for example, at least one of MATH0034, MATH0051, MATH0052 or MATH0053). Good coding skills are also desirable.
<i>Lecturer:</i>	Dr J Talbot, Prof R Hill

Course Description and Objectives

Lean is a programming language that can be used to interactively prove mathematical theorems. More precisely, when one types a proof into Lean, the computer checks that the proof is correct. Furthermore, when one has typed part of a proof, the computer determines at any stage what we have proved so far, and what is still to be proved. Lean also has various high level commands which can be used to prove simple statements automatically. One very appealing feature of Lean for mathematicians is the existence of a large and growing library of formalised mathematics: <https://leanprover-community.github.io/mathlib-overview.html>

The ultimate goal of this course is to provide students with a level of proficiency in Lean that will allow them to formalise (i.e. prove in Lean) theorems of undergraduate level pure mathematics.

Recommended Texts

- J Avigad, K Buzzard, RY Lewis, and P Massot, *Mathematics in Lean* https://leanprover-community.github.io/mathematics_in_lean/
- J Avigad, L de Moura, and S Kong, *Theorem Proving in Lean* https://leanprover.github.io/theorem_proving_in_lean/introduction.html
- H Macbeth, *Mechanics of Proof* <https://hrmacbeth.github.io/math2001/>

Detailed Syllabus

The course will be taught entirely through workshops during which students will learn by doing. We will focus on tactic mode proofs, and in particular on methods for automating proofs. We will not emphasise the theory of dependent types.

1. Basic introduction to type theory as required to use Lean, e.g. types, terms, Prop, equality (syntactic, definitional, propositional).
2. Introduction to Lean tactics.
3. Functions and Propositional Logic.
4. Sets and subtypes.

5. Inductive types.
6. Structures and classes.
7. Formalisation of selected theorems from real analysis, algebra, number theory, combinatorics and topology.

September 2026